Year 8 SEAL Science — Energy Practice Test (Medium)

TEACHER MARKING GUIDE — not for distribution to students

Total 40 marks: Section A 12 × 1 mark (separate answer sheet), Section B 28 marks across 7 questions. Writing time 55 min. Marking conventions: one mark per idea or step; no half marks. Mark units once, at the final answer. Where a later part uses an earlier answer, mark it consequentially. This is a practice paper at medium difficulty: two-step calculations, one unit conversion (W to kW in Q6), and a sentence starter only on Q3b.

Section A — Multiple-choice answer key (12 marks)

Key balance: A ×3, B ×3, C ×3, D ×3; no letter repeats on consecutive questions. Each wrong letter points to one misconception or slip, so a tally of wrong letters shows what to reteach.

Q	Key	Working for the key	Distractor diagnosis
1	B	A hay bale in the loft has gravitational potential energy, a stored type.	A: thermal energy is kinetic (moving particles), not stored. C: sound is kinetic (vibrating particles). D: chemical energy is potential (stored), not kinetic.
2	A	36,000 = ½ × 8,000 × v^2; v^2 = 9; v = 3.0 m/s	B: left out the 2 (√(36,000 ÷ 8,000)). C: KE ÷ m, no 2 and no square root. D: forgot the square root (v^2 = 9).
3	C	Hot-water bottle to lamb: thermal energy moves from one object to another and stays thermal.	A: chemical → kinetic and thermal. B: light → electrical → chemical. D: chemical → thermal and light. Each picks a change of type, confusing transformation with transfer.
4	D	Roll P is 70 degrees above the room, roll Q only 30 degrees; the bigger difference gives faster heat transfer.	A: heat as a stored substance; thinks cooler objects give heat away more easily. B: ignores the temperature difference with the room. C: 'cold' as a substance that flows in.
5	A	Wind KE → KE of blades and rod → GPE of lifted water	B: assumes every windmill makes electricity (a pump windmill is mechanical). C: mislabels moving air as stored energy. D: calls the wind chemical; water resting in a tank has GPE, not KE.
6	C	P 25%, Q 30%, R 20%; Q has the highest share useful	A: compares wasted joules (P 300 J), not the share. B: biggest useful output, ignoring the input. D: smallest input, ignoring the output.
7	B	Methods and materials suited local Country.	A: energy created by rubbing (conservation). C: ignores local knowledge and different Country. D: heat 'stored' in cold stones; sparks come from the strike.
8	D	1 kWh = energy of 1,000 W for 1 h	A: confuses energy with power. B: confuses kWh with time. C: confuses energy with cost.
9	A	Heater 3.6 × 4 = 14.4 kWh; cooler 2.0 × 10 = 20 kWh; pump 0.75 × 24 = 18 kWh	B: power alone, ignoring time. C: time alone, ignoring power. D: thinks energy use needs the star rating; it can be calculated from power × time.
10	C	Higher R-value = slower heat flow, both ways	A: reverses the R-value scale; insulators 'hold' heat. B: insulation only works one way. D: insulation 'makes' heat.
11	B	60,000 = 500 × 10 × h; h = 12 m	A: inverted the formula (5,000 ÷ 60,000). C: put ½ into GPE (mixed up with KE). D: left out g.
12	D	240 − 190 = 50 kWh; 50 × 30 c = 1,500 c = $15.00	A: kWh difference, not multiplied by the tariff. B: used 30 (cents) as dollars. C: added the kWh (430 kWh × 30 c) instead of subtracting.

Section B — Short-answer marking scheme (28 marks)

Read the Reject lists before marking so the class is marked consistently.

Question 1 — Hay season (7 marks)

a) (2 marks)

KE = ½ × 5,000 × 2.0² = 10,000 J

1 mark correct substitution into KE = ½mv^2. **1 mark** 10,000 J (unit needed).

Reject: 20,000 J (the ½ left out) or 5,000 J (speed not squared): no answer mark.

b) (2 marks)

GPE = 24 × 10 × 3.5 = 840 J

1 mark substitution into GPE = mgh. **1 mark** 840 J.

Reject: 84 J (g left out); 420 J (a ½ added).

c) (1 mark)

Efficiency = 840 ÷ 1,400 × 100% = 60%

1 mark 60%.

Consequential: On part b): accept (b answer) ÷ 1,400 × 100%.

Reject: 167% (1,400 ÷ 840): an efficiency over 100% is impossible.

d) (1 mark)

Thermal energy (heat) and sound energy.

1 mark two forms named: thermal (heat) and sound.

Reject: Only one form. Words that are not forms of energy, such as 'friction', 'lost energy' or 'waste'.

e) (1 mark)

Grease the chain and rollers to cut friction. Also accept a more efficient motor.

1 mark a change that cuts wasted energy (usually by reducing friction).

Reject: 'Lift the bale higher' or 'use a bigger motor' (neither raises the useful share). 'Make it go faster'.

Question 2 — Windmill water pump (2 marks)

– (2 marks)

One input arrow, 'kinetic energy of wind 1,500 J', splitting into: gravitational potential energy of water 480 J (useful, straight on), thermal energy 960 J and sound energy 60 J (bending away).

1 mark input and all three outputs labelled with energy type and value. **1 mark** widths roughly to scale: thermal about twice the useful arrow, sound much the thinnest, and the outputs together as wide as the input.

Reject: All arrows the same width. Old (ungreased) values. An input wider or narrower than 1,500 J.

Question 3 — Dairy hot water (4 marks)

a) (2 marks)

Water particles hit the copper particles on the inside of the pipe and make them vibrate faster.

These copper particles bump into their neighbours and pass the energy along through the wall (conduction), so the outside surface heats up. Copper is a metal, a good conductor, so this happens quickly.

1 mark hot water makes the copper particles at the inner surface vibrate faster (energy passed from water to copper). **1 mark** vibrating particles pass energy to neighbouring particles through the wall: conduction (accept 'metals are good conductors' as part of this mark only with the particle step).

Reject: "Heat goes through the copper" or "copper attracts heat" with no particles. "The particles expand" or "particles get hotter and bigger".

b) (2 marks)

Foil mainly reduces radiation; wrapped tightly with no air gap, it saves little. Foam holds air still in tiny cells: still air conducts poorly and can't flow, so foam cuts conduction and convection, which cause most of the pipe's heat loss. Foam is the better choice.

1 mark foil: reduces radiation only (accept 'reflects heat'), with little effect when wrapped tightly. **1 mark** foam: holds still air, so reduces conduction (and convection), making it the better choice.

Reject: 'Foil keeps the heat in' or 'foam blocks heat' with no type of heat transfer named. 'Foam soaks up the heat and stores it'.

Question 4 — Fire plough (3 marks)

a) (2 marks)

Chemical energy in the ranger's muscles → kinetic energy of the moving stick. Friction in the groove transforms kinetic energy into thermal energy, making hot dust that forms the ember.

The ember's heat ignites the tinder. The tinder's chemical energy is transformed into thermal (heat) and light energy in the flame.

1 mark chemical (muscles) → kinetic (stick) → thermal by friction, forming the ember. **1 mark** in the flame, the tinder's chemical energy → thermal and light.

Reject: "Friction creates heat" with no energy types. "The ember's energy becomes the flame" without the tinder's chemical energy. Any answer that says energy is created.

b) (1 mark)

Thermal energy is transferred from the flames or hot coals to the food, by radiation, convection (hot air) or conduction (through a hot stone or the pan).

1 mark thermal energy moves from the fire to the food, with a method or path named.

Reject: 'The fire cooks the food' with no energy named. A transformation (e.g. chemical → thermal in the wood) given instead of a transfer to the food.

Question 5 — Farmhouse bedroom (2 marks)

a) (1 mark)

Convection (a draught: moving air carries heat; cold air comes in and pushes warm air out).

1 mark convection (accept 'draught' only if described as moving air).

Reject: Conduction; radiation.

b) (1 mark)

Double glazing: the gas or vacuum gap between the panes slows conduction. Also accept close-fitting heavy curtains (still air layer) or a low-E film (reflects radiant heat).

1 mark an upgrade AND how it reduces a named type of heat transfer.

Reject: "Put in a better window" with no detail. "Double glazing makes the glass thicker" (the gap does the work). An upgrade for the door gap (does not answer the question).

Question 6 — Country bakery (4 marks)

a) (2 marks)

9,000 W = 9 kW; energy = 9 × 6 = 54 kWh

Cost = 54 × \$0.28 = \$15.12 per day

1 mark 54 kWh (W converted to kW, or 54,000 Wh ÷ 1,000). **1 mark** \$15.12.

Consequential: Cost mark consequential on the kWh value.

Reject: 54,000 kWh (W not converted to kW) loses the first mark; mark the cost consequentially.

b) (2 marks)

Cost per hour = 9 × $0.28 = $2.52; $10.00 ÷ $2.52 = 3.97 h, so 3 whole hours (4 h would cost $10.08).

1 mark cost per hour ($2.52) or the energy allowed (10 ÷ 0.28 = 35.7 kWh). **1 mark** 3 hours, with the reason that 4 hours goes over $10.00.

Consequential: Consequential on a wrong per-hour cost carried from a).

Reject: "4 hours" (rounded 3.97 up without checking the cost); 3.97 h (not whole hours).

Question 7 — Solar panel investigation (6 marks)

a) (2 marks)

Independent: tilt angle. Dependent: power output (W). Kept the same (any one): same panel; facing north; same time of day; clear sky; same meter; same unshaded spot.

1 mark independent AND dependent both correct. **1 mark** one valid controlled variable.

Reject: "The sun" or "the weather" alone (too vague); "the angle" as a controlled variable; "the trials".

b) (2 marks)

Outlier: 30°, Trial 2, 5.2 W (far below 8.0 and 8.2 W). Average without it: (8.0 + 8.2) ÷ 2 = 8.1 W

1 mark identifies 5.2 W at 30° (Trial 2). **1 mark** 8.1 W.

Reject: 7.1 W (included the outlier). Identifying 75° or 0° values as the outlier.

c) (2 marks)

Not supported. Power rises from 6.7 W at 0° to 8.1 W at 30°, then falls to 5.9 W at 75°. Steeper is better only up to about 30°.

1 mark judgement that the conclusion is not (fully) supported, with data quoted from both sides of the peak. **1 mark** a corrected statement: power increases up to about 30°, then decreases (a best angle exists). Accept 'best between 15° and 45°' or 'about 30°'.

Consequential: Mark consequentially on the student's 30° average from b).

Reject: "Correct" (the data falls after 30°). A judgement with no data quoted.

Mark summary

Section	Marks	Notes
A — multiple choice	12	Answer sheet; key A3 / B3 / C3 / D3
B Q1 — Hay season	7	a 2, b 2, c 1, d 1, e 1
B Q2 — Windmill water pump	2	drawing
B Q3 — Dairy hot water	4	a 2, b 2
B Q4 — Fire plough	3	a 2, b 1
B Q5 — Farmhouse bedroom	2	a 1, b 1
B Q6 — Country bakery	4	a 2, b 2
B Q7 — Solar panel investigation	6	a 2, b 2, c 2
Total	40	

Test blueprint

Medium difficulty: comparable demand to a full unit test, with every context set on a farm or in a country town.

Demand level	Marks	Share	Medium target	Items
L1 Recall	8	20%	about 20%	MC 1, 7, 8, 10; SB 1d, 5a, 7a
L2 Routine procedure	12	30%	about 30%	MC 2, 11, 12; SB 1a, 1b, 1c, Q2 (labels), 6a, 7b (average)
L3 Interpret or explain	15	38%	about 35%	MC 3, 4, 5, 6, 9; SB Q2 (widths), 3a, 4a, 4b, 5b, 6b, 7b (outlier)
L4 Evaluate	5	13%	about 15%	SB 1e, 3b, 7c

Coverage by unit lesson

Unit lesson	Marks	Items
2 Forms of energy, KE and GPE	8	MC 1, 2, 5, 11; SB 1a, 1b
3 Heat transfer	5	MC 4; SB Q3
4 Transfers, transformations, efficiency, Sankey diagrams	7	MC 3, 6; SB 1c, 1d, 1e, Q2
5 First Nations fire knowledge and uses of fire	4	MC 7; SB Q4
6 Appliances, star ratings, running cost	6	MC 8, 12; SB Q6
7 Household energy design	3	MC 10; SB Q5
8 Investigation skills	6	SB Q7
9 Energy audit	1	MC 9

55 minutes for 40 marks gives about 1.4 minutes a mark. The slowest items are SB Q2 (Sankey drawing) and SB 6b (working back from a budget).